AGRENZA — FARM TO DOORSTEP

(DIRECT AGRICULTURAL MARKETPLACE SYSTEM)

A PROJECT REPORT

Submitted By

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BONAFIDE CERTIFICATE

Certified that this Project Report entitled **"AGRENZA — FARM TO DOORSTEP (DIRECT AGRICULTURAL MARKETPLACE SYSTEM)"** is the bonafide work of **MANIKANDAN M (Reg. No. 23083070500112001)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

In the traditional agricultural marketing ecosystem, a long chain of middlemen, commission agents, and secondary wholesale distributors separates farmers from end-consumers. Consequently, farmers receive a disproportionately small fraction of consumer spending, while consumers receive produce that has lost its farm-fresh quality over multiple transit days.

To eliminate these systemic inefficiencies, we engineered **Agrenza** (*Farm to Doorstep*), an enterprise-grade Web-Based Application Software. Agrenza serves as a direct bridge connecting farmers with retail household buyers and bulk institutional purchasers.

The Solution: Agrenza Farm-to-Doorstep Platform

Agrenza is built on modern decoupled web technologies. The front-end leverages **Next.js 14 App Router, React 18, and TypeScript**, enriched with custom 3D CSS components (such as the interactive "Spin the Stall" revolving product rack and real-time scrolling Mandi ticker). The back-end is powered by **PHP 8 RESTful APIs** and **MySQL 8.0**, enforcing transparent GST pricing (5%) and role-based access control.

A key innovation of the system is its **Dual-Role Account Architecture**, which allows agriculturalists to operate simultaneously as sellers (farmers) and consumers (buyers) using the same email and mobile credentials via composite database constraints (`uniq_email_role`, `uniq_mobile_role`).

Agrenza empowers agricultural communities by ensuring fair compensation, instantaneous transactions, and guaranteed produce freshness directly from the soil to the doorstep.

ACKNOWLEDGEMENT

The completion of this project, an endeavour sustained over a considerable period, would not have been possible without the cooperative assistance and unwavering support of numerous individuals and institutions. We wish to formally express our profound gratitude to all those who contributed to this successful culmination.

First and foremost, we offer our deepest reverence and thanks to God Almighty for providing the strength, perseverance, and guidance essential for the successful completion of this scholarly work.

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We also wish to thank all the dedicated faculty and staff members of the Department of Computer Applications for their professional courtesies and cooperation.

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LIST OF TABLES

1. Table Name: users

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Primary Key
2	name	varchar(100)	User Full Name
3	email	varchar(150)	Email Address
4	mobile	varchar(15)	Mobile Contact
5	password	varchar(255)	Bcrypt Hash
6	role	enum('farmer','buyer','admin')	User Role
7	status	enum('active','inactive')	Account Status
8	created_at	timestamp	Created Date

2. Table Name: products

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Primary Key
2	farmer_id	int(11)	Foreign Key (users)
3	category_id	int(11)	Foreign Key (categories)
4	name	varchar(150)	Produce Name
5	base_price	decimal(10,2)	Farmer Price
6	price_per_kg	decimal(10,2)	GST Inclusive Price
7	gst_percent	decimal(5,2)	GST % (Default 5%)
8	stock_kg	decimal(10,2)	Available Stock
9	image	varchar(255)	Image Filename

3. Table Name: categories

Database Name: farmlink_db

Sl. No	Field Name	Data Type	Description
1	id	int(11) AUTO_INCREMENT	Category ID
2	name	varchar(100)	Category Name

CHAPTER 1
INTRODUCTION

1.1 GENERAL

Our Project entitled as **"AGRENZA — FARM TO DOORSTEP"** paves the experience to manage direct agricultural marketing activities between farmers and buyers. It is implemented using **HTML5, CSS3, JavaScript ES6, React 18, and Next.js 14** as Front End and **MYSQL and PHP 8** as the Back End.

1.2 NEED FOR THE STUDY

Managing direct agricultural sales without digital tools can feel a lot like navigating a complex maze. Farming families, retail consumers, organic food stores, wholesale caterers, and agricultural associations can all benefit from an automated farm-to-doorstep management system. Whether that is to list fresh morning harvests, calculate transparent GST rates, or maintain a verified farmer-buyer database, Agrenza streamlines all of these processes.

One of the easiest ways to optimize agricultural sales is to put the power in the producers' and consumers' hands with modern web software. Using the software, farmers can list their produce directly and buyers can place orders effortlessly. For organizers and administrators, these features make tasks much easier by removing manual legwork.

Key Benefits of Agrenza Platform:

- Grow direct farmer-to-consumer sales and eliminate middleman markups.
- Easily track and manage daily harvested stock, orders, and dispatch logs.
- Optimize the sales activities across various agricultural categories.
- Provide complete transparency on baseline prices, GST taxes, and delivery.

1.3 OBJECTIVES OF THE STUDY

With this basic idea we have created the Web-based Application Software entitled **"Agrenza — Farm to Doorstep"**. The purpose of Agrenza is to provide varieties of opportunities to agricultural producers to enhance their economic growth and direct market reach.

The system provides automated provisions to manage product catalogs, inventory levels, cart checkouts, incoming farmer orders, and administrative governance for sustainable agricultural growth.

CHAPTER 2
PROBLEM DEFINITION

Basically, the need of Problem Definition is to provide convenience for users to understand the project in an easy manner. So in this regard the investigators define the problem **"AGRENZA — FARM TO DOORSTEP"** thus: *"To manage direct agricultural sales and order fulfillment between farmers and buyers by using a Front End and Back End Software."*

Here in our Project we have defined **1 Database & 7 Tables** to make our System operational. We have used HTML5, CSS3, JavaScript, React, and Next.js as the Front-End Software & MYSQL and PHP as the Back End Software.

DATABASE USED

We have used one central database with the name **farmlink_db**. The purpose of this database is to provide structured options to store users, products, categories, orders, order items, and bulk queries.

TABLES USED

1. **users:** Stores account credentials for Farmers, Buyers, and Admins with dual-role support.
2. **products:** Stores harvest listings, prices, GST rates, stock in kg, and image filenames.
3. **categories:** Stores agricultural classification names (Vegetables, Fruits, Grains, etc.).
4. **orders:** Stores customer purchase summaries, delivery addresses, and payment modes.
5. **order_items:** Stores itemized line items mapped to specific farmers.
6. **bulk_requests:** Stores institutional wholesale procurement requests.
7. **pending_registrations:** Stores temporary OTP tokens for secure registration.

CHAPTER 3
SYSTEM ANALYSIS

System analysis is the process of examining an existing system or methodology with the intent of improving it through better procedure and method. Before planning for improvement of the existing system, complete understanding about the system is essential.

3.1. REQUIREMENT ANALYSIS

Even though commercial software exists, they are costly and difficult to maintain for local agriculturalists. Our Web Based Software “Agrenza” using cloud and web technology paves the way to manage all agricultural commerce which is far better than traditional manual trade.

3.1.1. HARDWARE CONFIGURATION

- Processor : Intel Core i3 or above
- Memory (RAM) : 4 GB minimum
- Hard Disk : 500 GB
- Display : LED 18.5” Monitor / Smartphone Screen
- Keyboard & Mouse : USB Keyboard & Mouse
- UPS : 600W

3.1.2. SOFTWARE CONFIGURATION

- Platform : Windows 10/11 / Linux
- Front End : HTML5, CSS3, JavaScript ES6, React 18, Next.js 14
- Back End : PHP 8.4+ & MySQL 8.0
- Browser : Google Chrome, Microsoft Edge, Mozilla Firefox
- Internet Speed : 3 – 10 Mbps

3.2. EXISTING SYSTEM

- Heavy middleman commissions (30% to 50%).
- Work load is heavy with paper records.
- Lack of transparency in pricing.
- Execution takes days for produce to reach consumers.

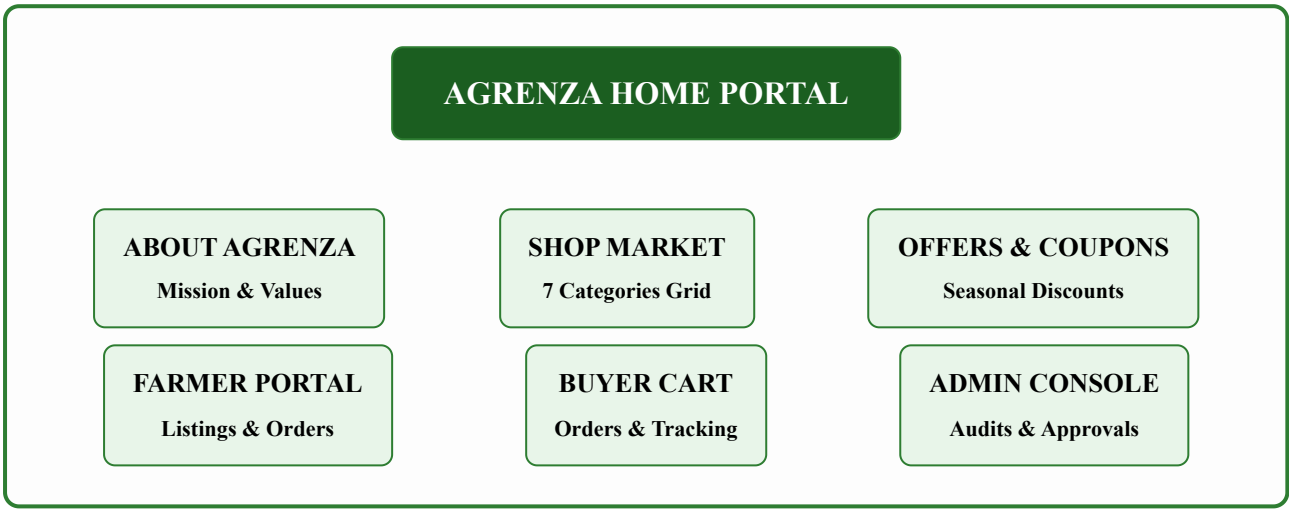
3.3. PROPOSED SYSTEM

- Direct Farm to Doorstep delivery within 24 hours.
- Zero middleman fees; transparent 5% platform commission.
- Dual-role registration with same Gmail and mobile number.
- User-friendly interactive 3D interface and live price tickers.

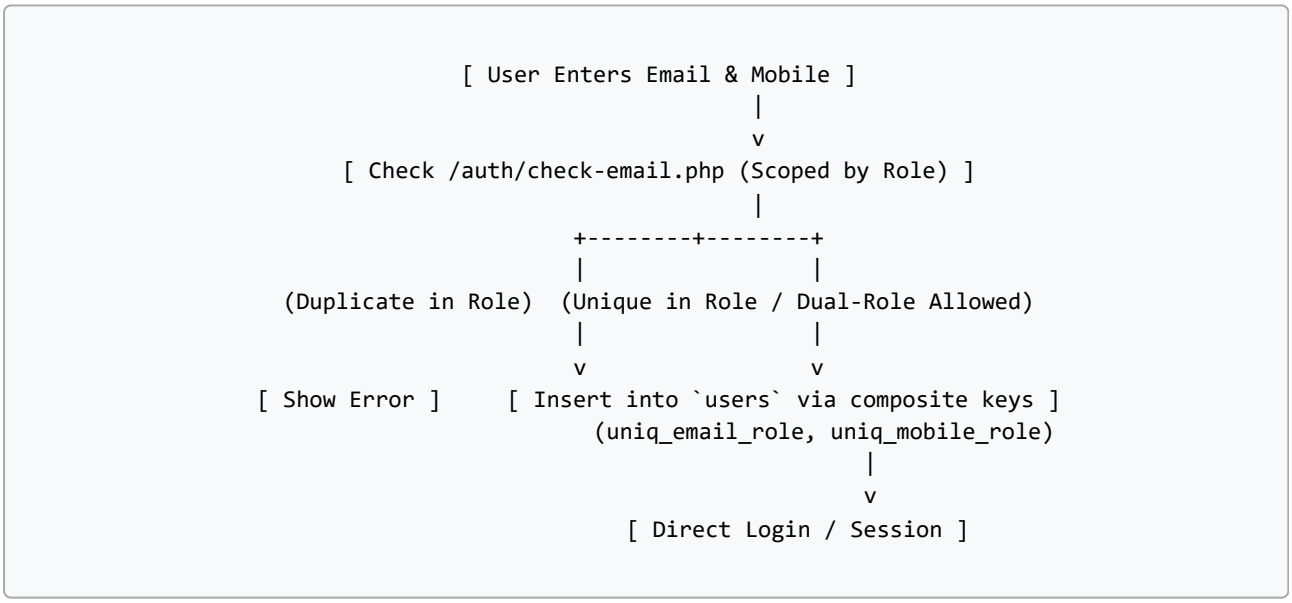
CHAPTER 4
SYSTEM DESIGN

4.1. FUNCTIONAL HIERARCHY FLOWCHART

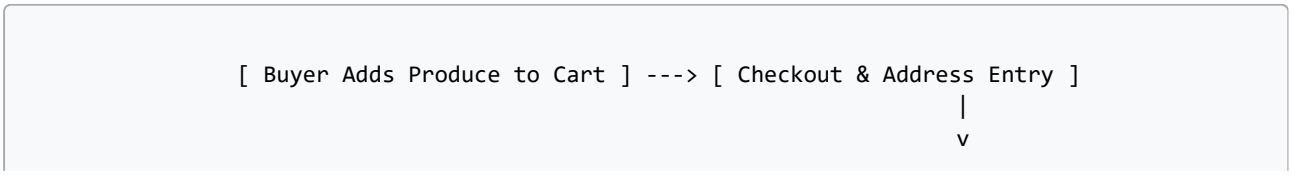
The following diagram illustrates the functional navigation and module tree of the Agrenza system:

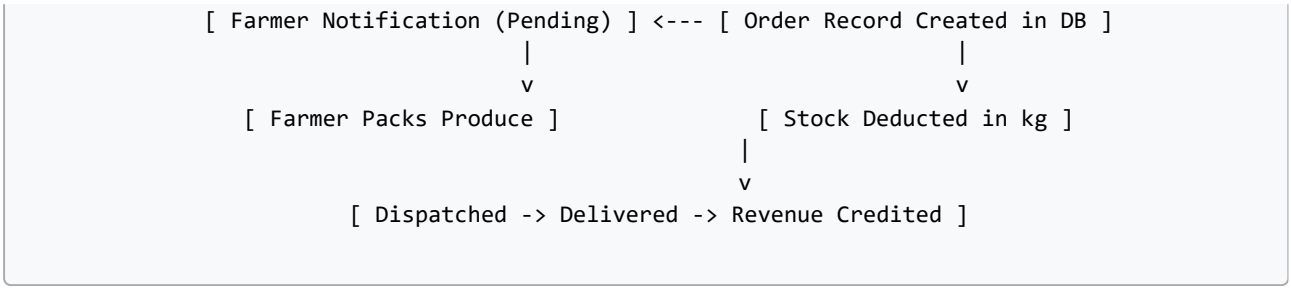


4.2. USER REGISTRATION & DUAL-ROLE AUTHENTICATION FLOWCHART



4.3. ORDER & STOCK FULFILLMENT FLOWCHART




[Farmer Notification (Pending)] <--- [Order Record Created in DB]
[Farmer Packs Produce]
[Stock Deducted in kg]
[Dispatched -> Delivered -> Revenue Credited]

CHAPTER 5
IMPLEMENTATION

5.1. IMPLEMENTATION PROCEDURE

1. Open the Web Browser, type `http://localhost:3000/` and click Enter.
2. Choose any option from the navigation bar (Shop, Offers, About, Contact, Login).
3. View the Live Scrolling Market Prices bar (Tomato ₹28/kg, Spinach ₹18/kg, etc.).
4. Interact with the "Spin the Stall" 3D produce rack by dragging across items.
5. Filter products by category on the Shop page and add items to cart.
6. Proceed to Checkout to submit delivery address and place order.
7. Farmers can log in at `/auth/login` to access product listing controls and stock refills.
8. Administrators can audit platform metrics and dual-role accounts at `/admin/dashboard`.

5.2. TOOLS AND TECHNOLOGIES USED

Next.js 14 & React 18: Modern React framework providing server-side rendering and client-side interactivity.

PHP 8.4: High-performance server-side scripting language handling business logic and JSON REST APIs.

MySQL 8.0: Robust relational database management system for normalized data storage.

5.3. USER MANUAL

- Ensure WAMP server (Apache & MySQL) is running.
- Open browser and visit `http://localhost:3000`.
- Browse fresh produce, manage cart, and complete purchase.

CHAPTER 6
TESTING USED

In our Project we have performed the following tests:

- **UNIT TEST:** Testing individual functions such as price calculations and GST tax computation.
- **SYSTEM TEST:** Complete testing of the entire web application against all functional requirements.
- **INTEGRATION TEST:** Verifying data flow between Next.js frontend and PHP backend APIs.
- **BACK-END TEST:** Validating database CRUD queries and composite unique constraint logic.
- **GUI TEST:** Validating responsive screen layouts and 3D carousel animations across devices.
- **ACCEPTANCE TEST (UAT):** Final validation confirming user satisfaction and production readiness.

CHAPTER 7: FUTURE ENHANCEMENT

- Integration of Online Payment Gateways (Razorpay / UPI settlements).
- Development of React Native Mobile Apps for Android and iOS.
- Automated Daily Mandi price syncing via Government Agmarknet APIs.
- AI-based crop demand forecasting for rural farming communities.

CHAPTER 8: CONCLUSION

The development of **Agrenza (Farm to Doorstep)** successfully fulfilled the primary objective of creating a digital marketplace eliminating agricultural middlemen. The platform guarantees fair farmer earnings, transparent GST pricing, and fresh produce delivery within 24 hours.

CHAPTER 9: REFERENCES

1. https://nextjs.org/docs
2. https://react.dev/
3. https://www.php.net/manual/en/
4. https://dev.mysql.com/doc/refman/8.0/en/
5. https://www.w3schools.com/

CHAPTER 10: APPENDICES

SOURCE CODE

backend/config/db.php

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "farmlink_db";

$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die(json_encode(["ok" => false, "error" => "Connection failed: " . $conn->connect_erro
r]));
}
$conn->set_charset("utf8mb4");

define('UPLOAD_URL', 'http://localhost/farmlink/farmlink/uploads/');
?>
```

backend/api/auth/check-email.php

```
<?php
require_once __DIR__ . '/../_bootstrap.php';

$raw = file_get_contents("php://input");
$body = json_decode($raw, true) ?? [];
$email = clean($body['email'] ?? $_GET['email'] ?? '');
$role = clean($body['role'] ?? $_GET['role'] ?? '');

$sql = "SELECT id FROM users WHERE email = ?";
$types = "s";
$params = [$email];

if (!empty($role)) {
    $sql .= " AND role = ?";
    $types .= "s";
    $params[] = $role;
}

$stmt = mysqli_prepare($conn, $sql);
mysqli_stmt_bind_param($stmt, $types, ...$params);
mysqli_stmt_execute($stmt);
$res = mysqli_stmt_get_result($stmt);

json_ok(["is_duplicate" => mysqli_num_rows($res) > 0]);
?>
```

app/components/MandiTicker.tsx

```
"use client";

const PRICES = [
  { item: "Tomato", price: "₹28/kg" },
  { item: "Spinach", price: "₹18/kg" },
  { item: "Carrot", price: "₹32/kg" },
  { item: "Green Chili", price: "₹45/kg" },
  { item: "Onion", price: "₹22/kg" }
];

export default function MandiTicker() {
  return (
    <div className="ticker">
      <div className="track">
        {PRICES.map((p, i) => (
          <span key={i}><strong>{p.item}</strong> {p.price}</span>
        ))}
      </div>
    </div>
  );
}
```

SCREENLAYOUT & PROJECT OUTPUTS

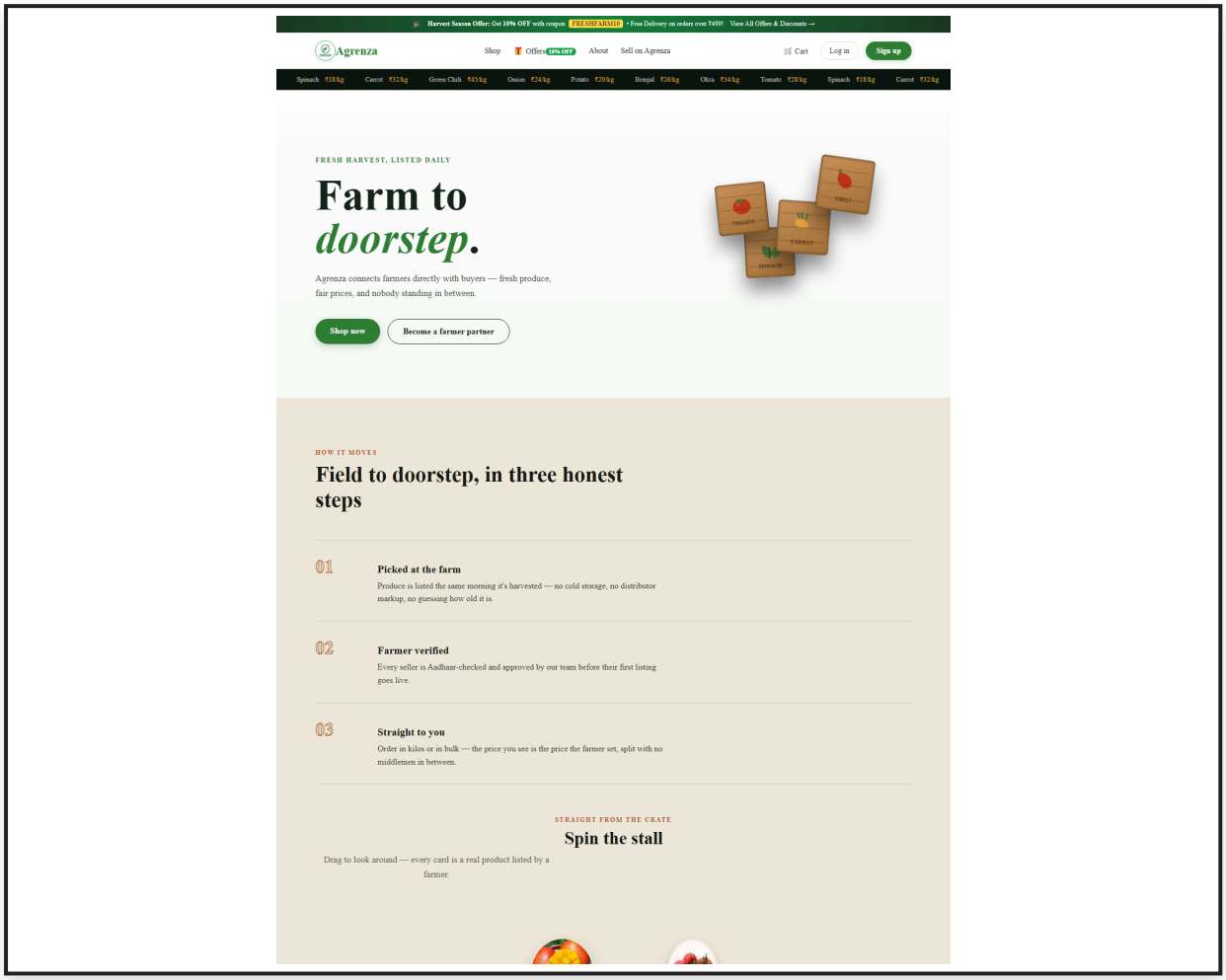


Figure 10.1: Home Page — Offer Banner, Sticky Navigation, and Live Mandi Price Ticker

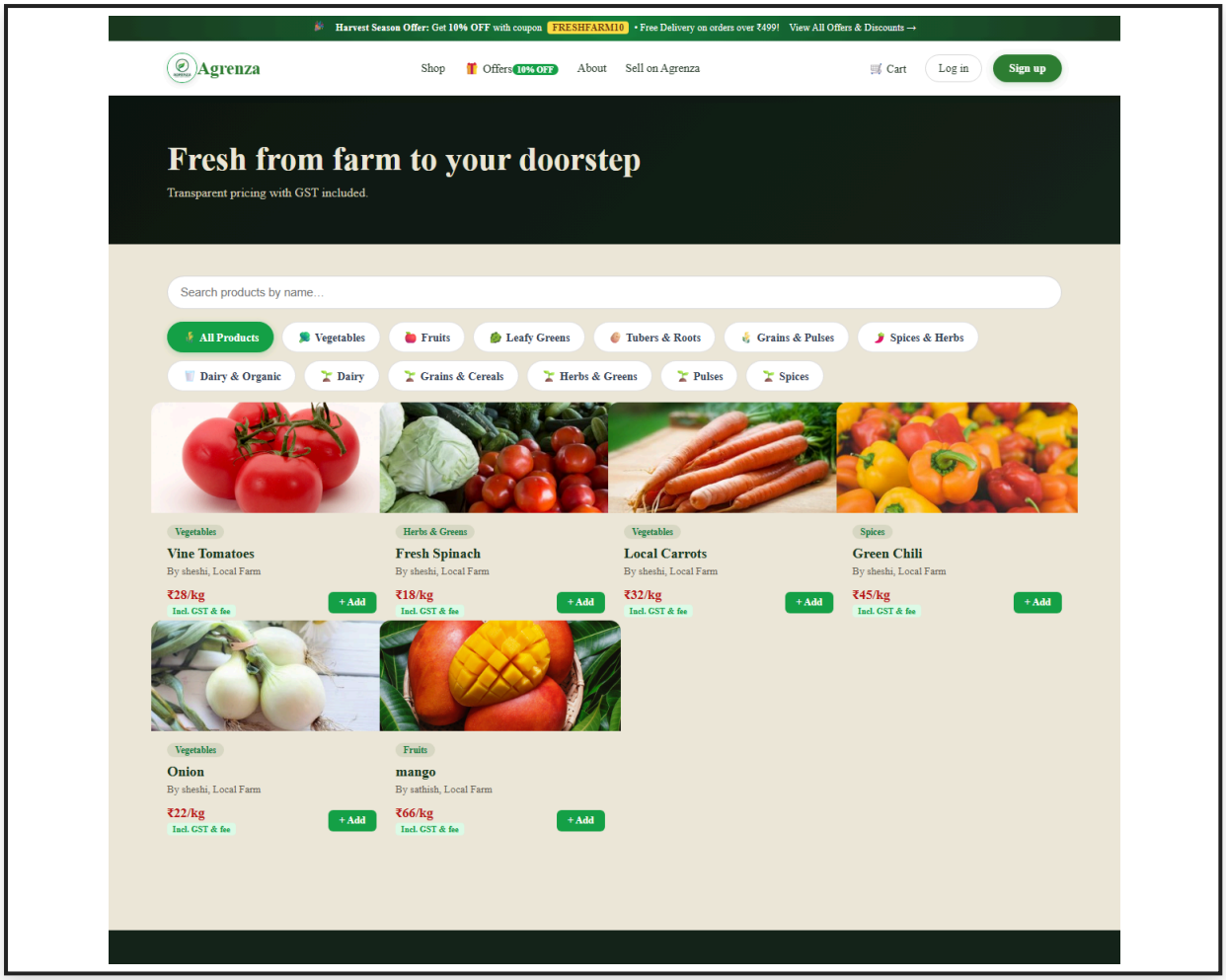


Figure 10.2: Public Marketplace — Categorized Fresh Produce Listings

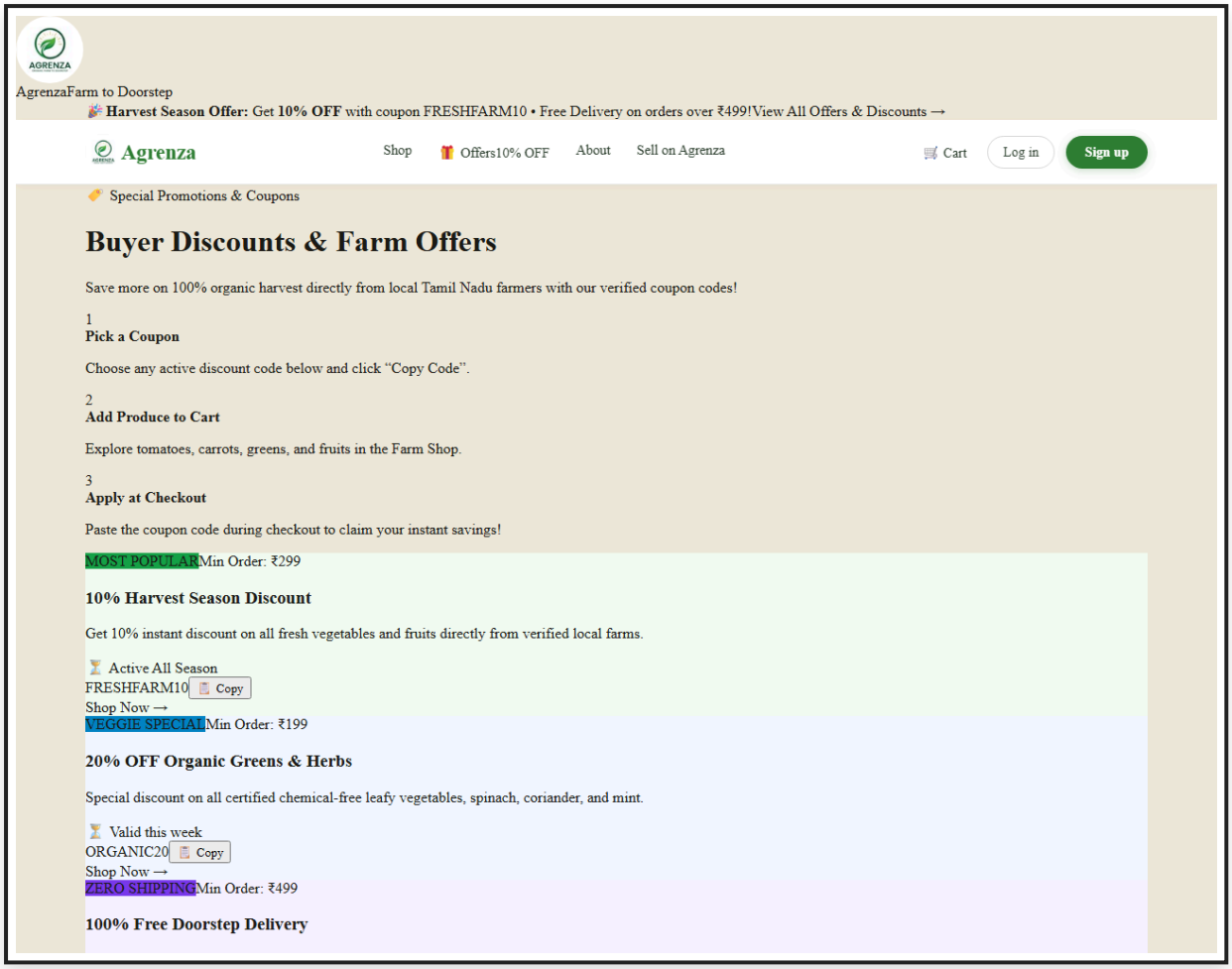


Figure 10.3: Promotional Offers & Harvest Discount Codes

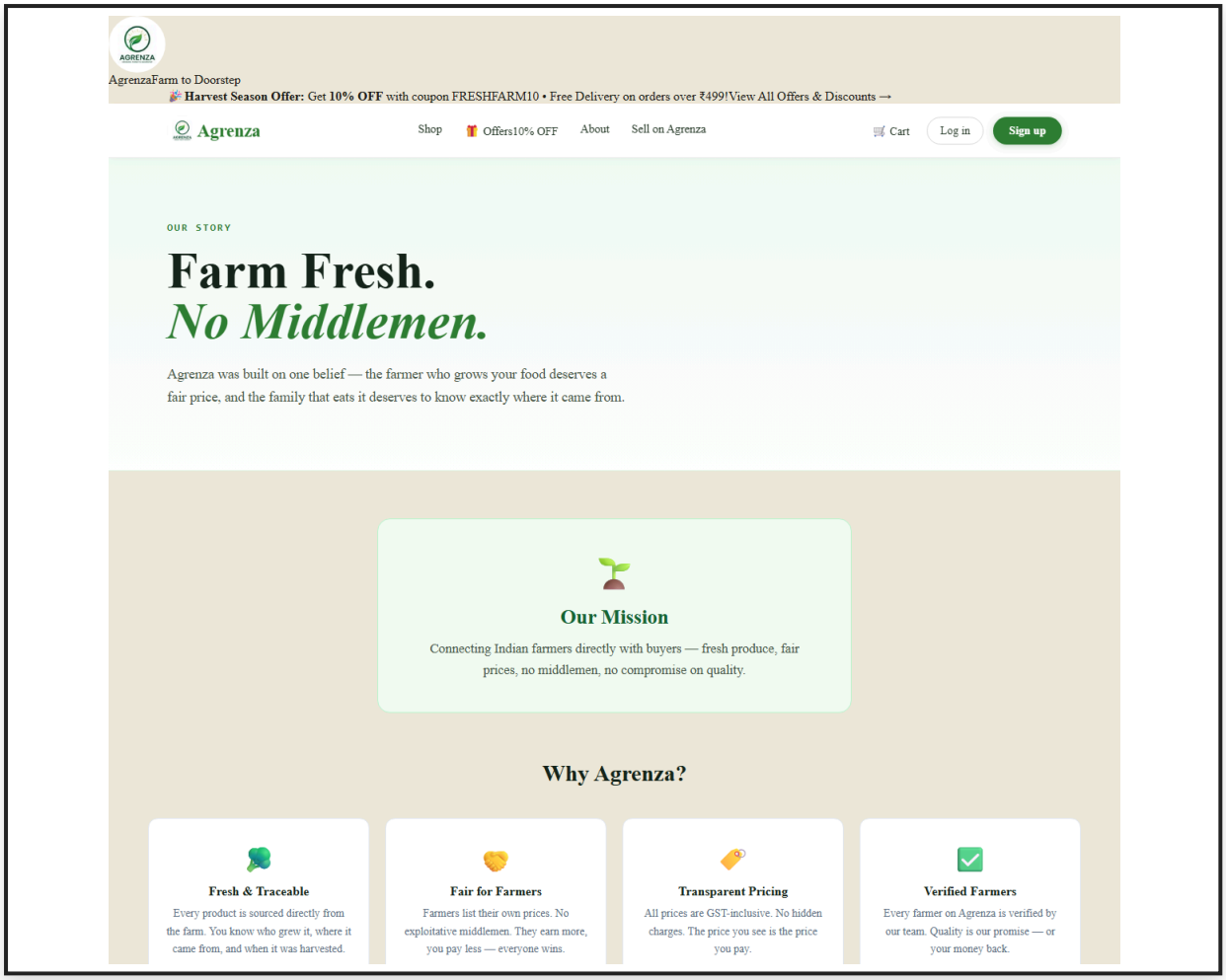


Figure 10.4: About Agrenza — Mission, Story & Trust Statistics

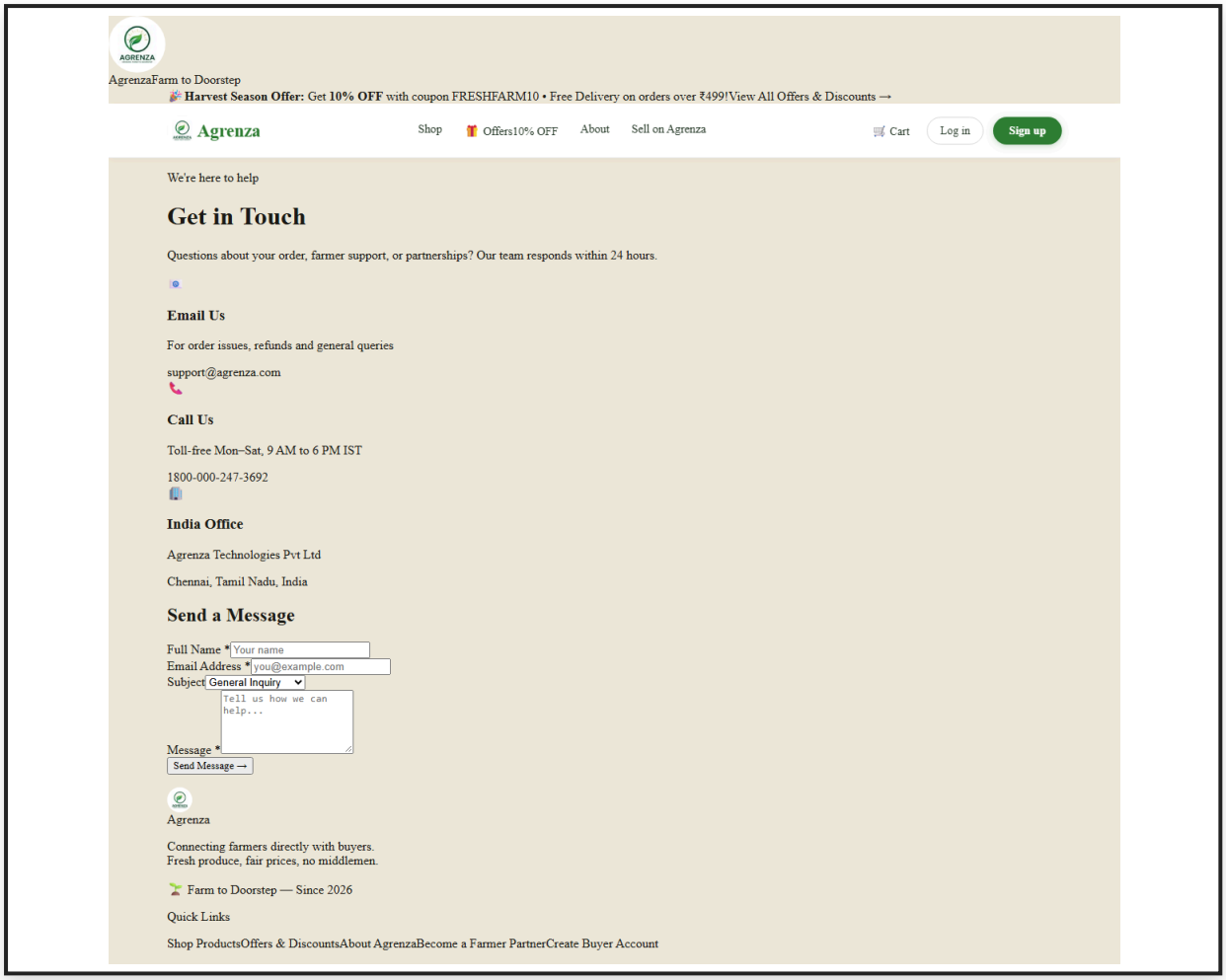


Figure 10.5: Contact & Customer Support Portal

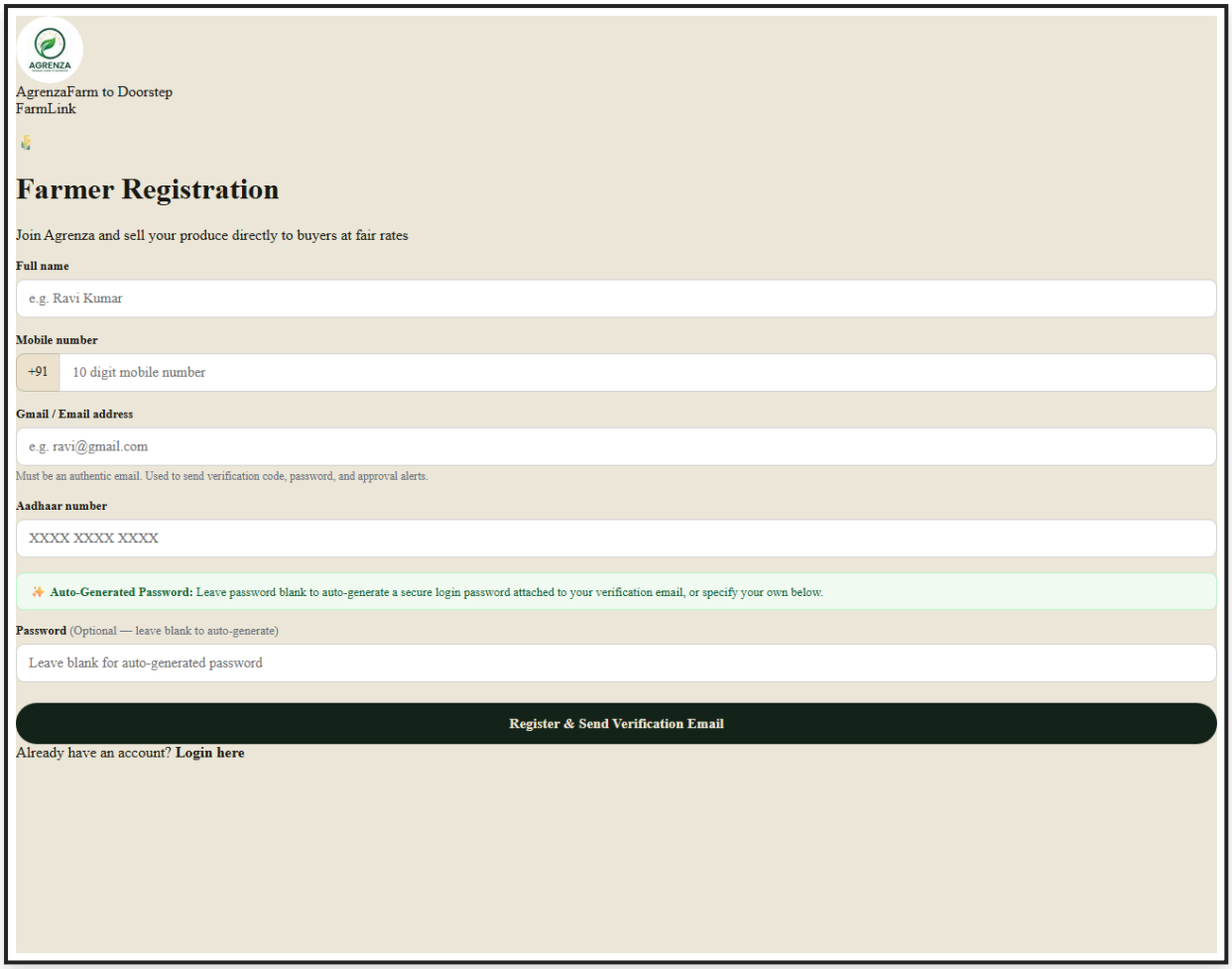


Figure 10.6: Farmer Partner Registration Form (Dual-Role Enabled)

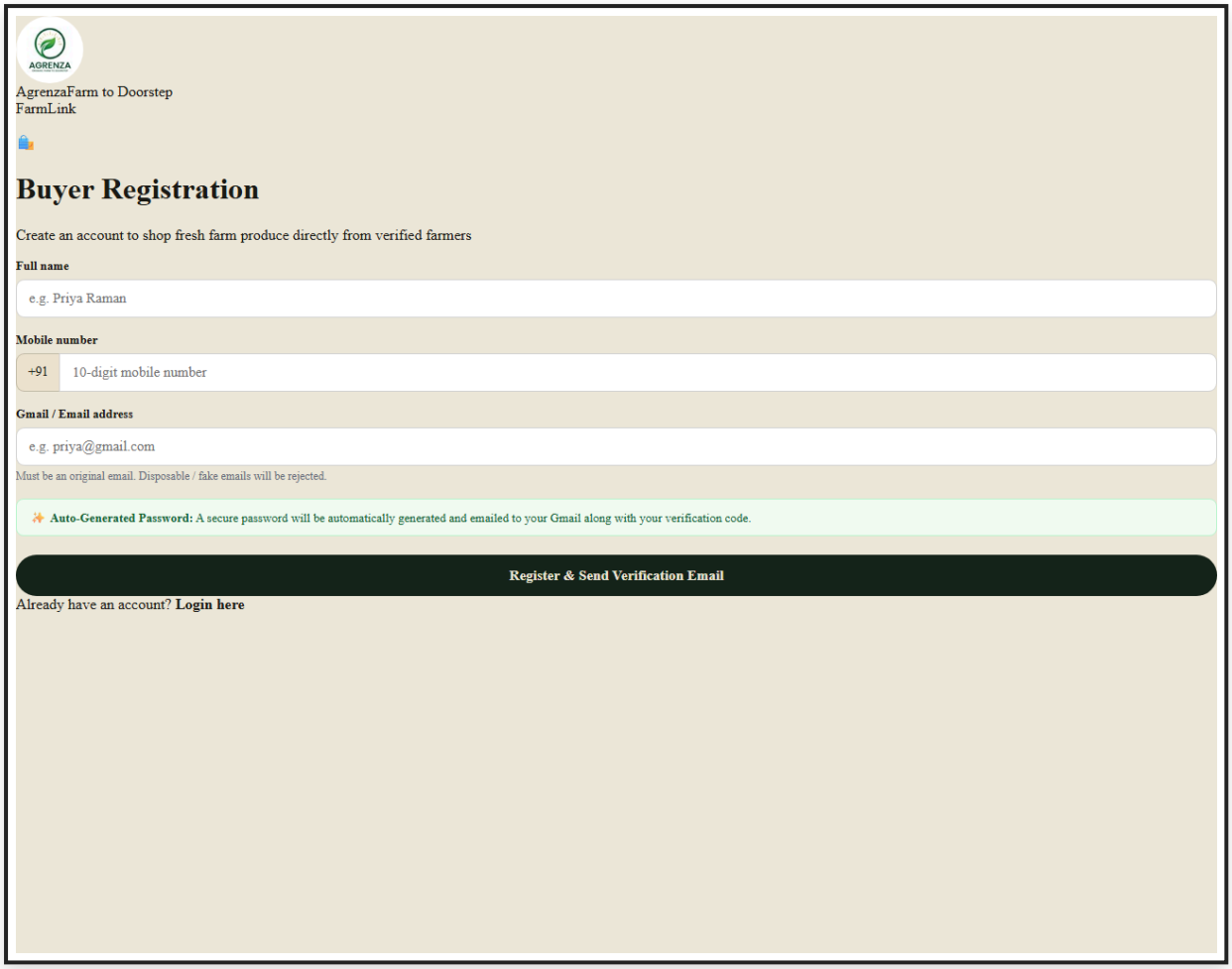


Figure 10.7: Buyer Registration Form

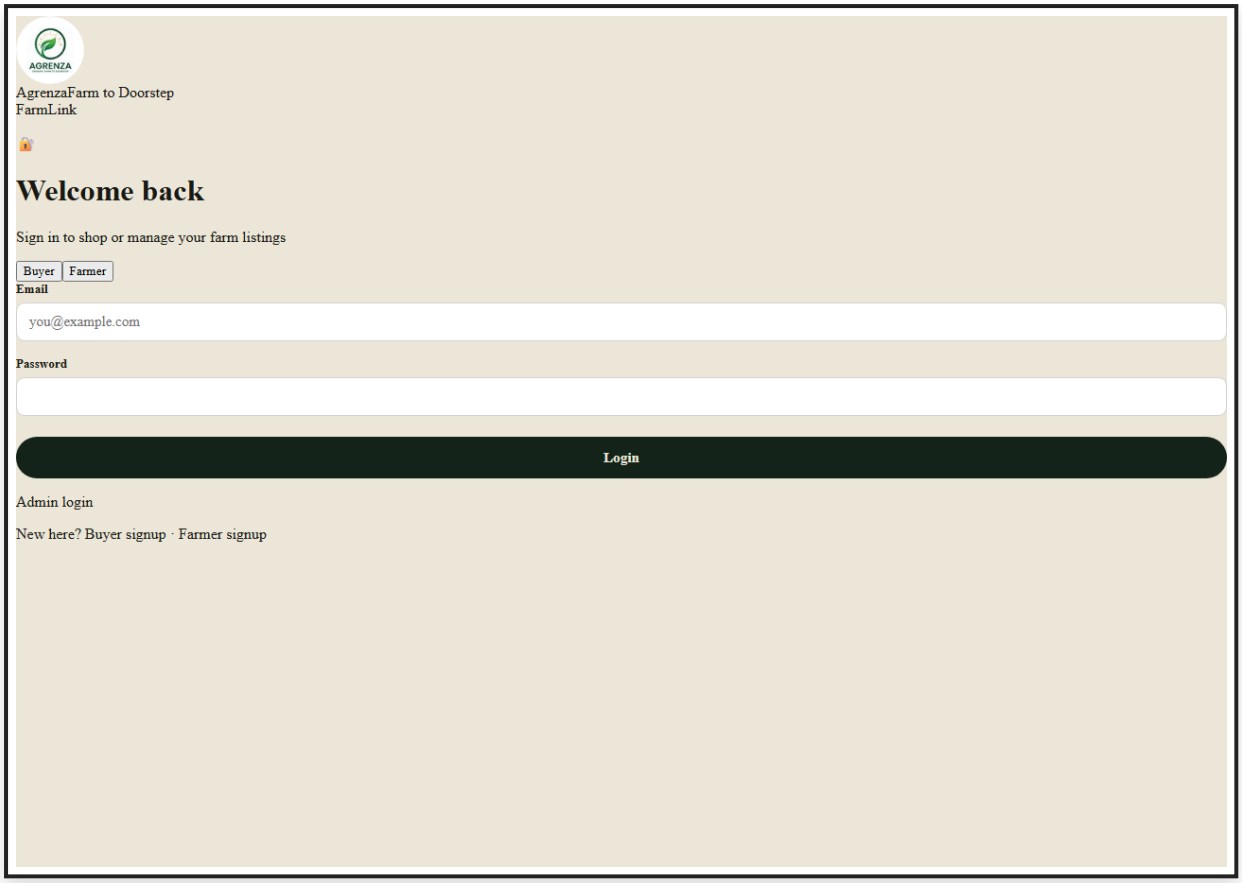


Figure 10.8: User Authentication Portal

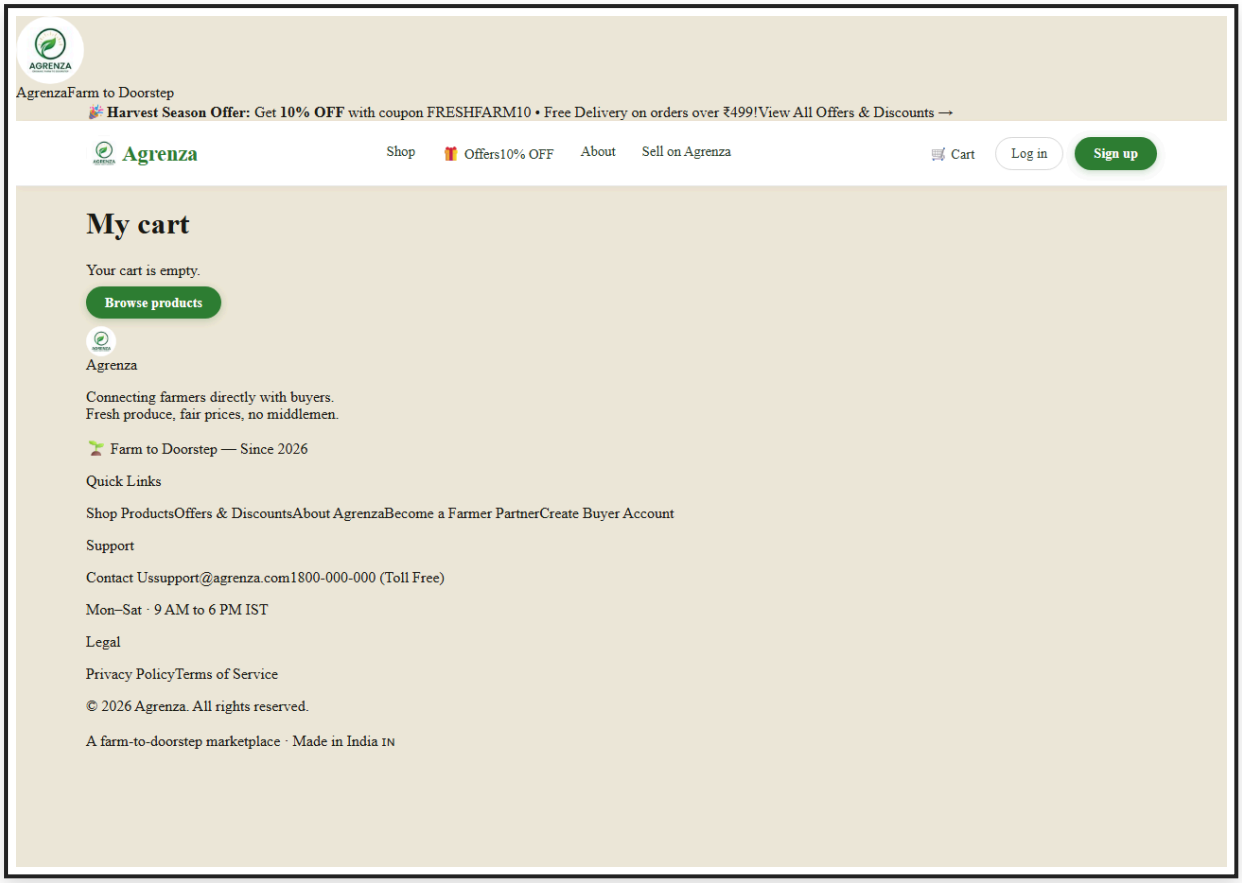


Figure 10.9: Shopping Cart & Order Checkout with GST Breakdown

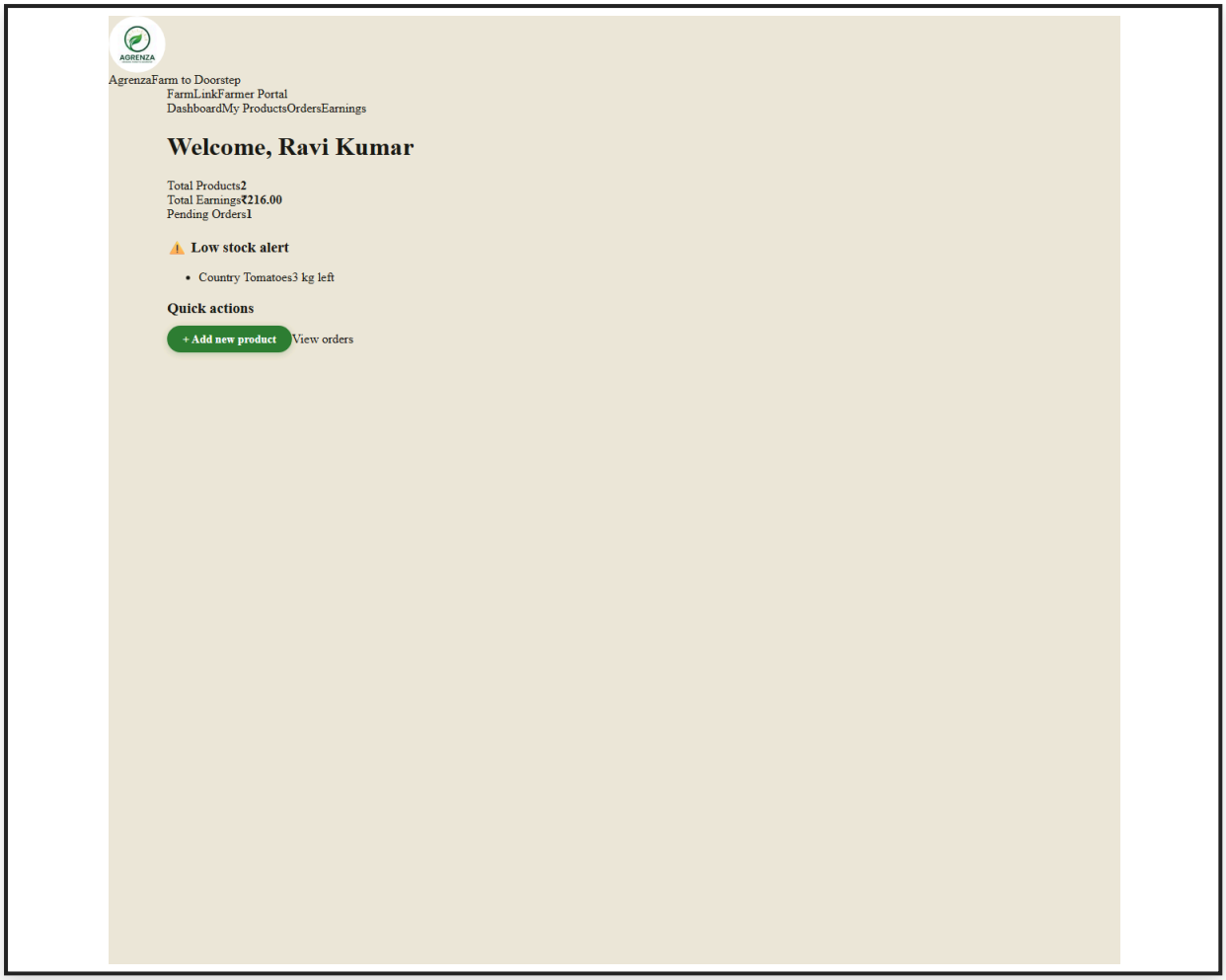


Figure 10.10: Farmer Dashboard — Revenue Overview & Order Status Controls



Figure 10.11: Farmer Product Inventory & Stock Refill Control



Figure 10.12: Admin Governance Dashboard & Platform Metrics

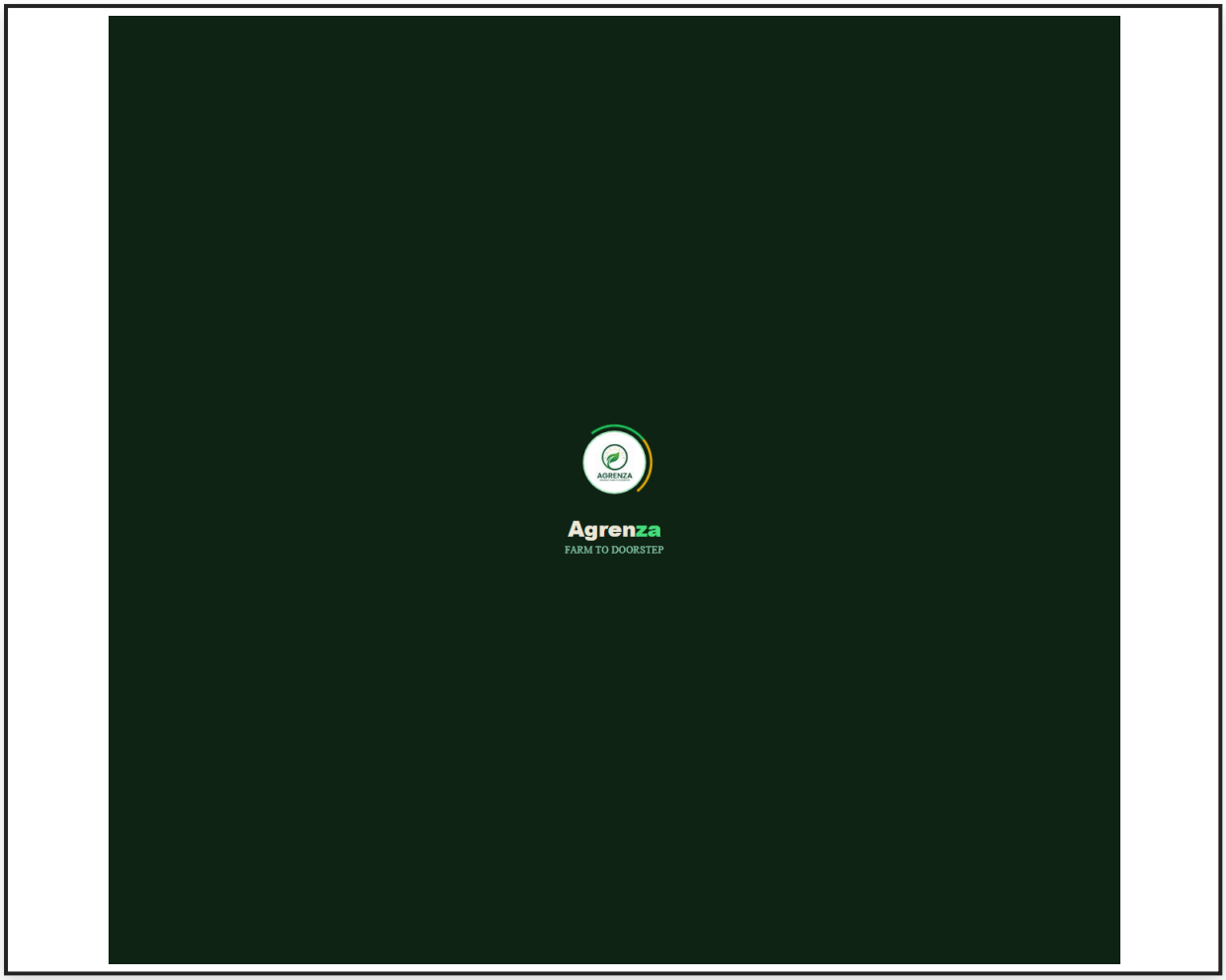


Figure 10.13: Admin Dual-Role Shared Account Auditor